Pebble-scale topology effects on homogenised neutronics of Li_4SiO_4/Be breeder beds

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Abstract

Volume-homogenised breeder-bed models remove the connected He void topology that controls neutron free paths, first-collision ordering and fast-neutron exposure of Be multiplier pebbles. This study compares material-equivalent homogenised and geometry-resolved Li_4SiO_4/Be representative volume elements (RVEs) in OpenMC using strict non-overlapping hard-sphere packings. For the audited baseline at packing fraction (PF) = 0.618, with reflective lateral boundaries and axial vacuum boundaries, the resolved geometry increases the local ^{6}Li(n,t) tritium-production response (TPR_6) by $+17.47 \pm 0.71$ %, total H3-production by $+29.13 \pm 0.25$ % and ^{9}Be(n,2n) by $+45.80 \pm 0.11$ %, while reducing the OpenMC neutron-mode heating score by -18.49 ± 0.04 % and increasing track-length flux by $+23.08 \pm 0.09$ %. A PF sweep from 0.56 to 0.618 shows decreasing topology-induced shifts with increasing PF, and three independent PF=0.618 realisations preserve the sign and order of the bias. A 50-group spectrum diagnostic gives a resolved/homogenised flux ratio of 1.221 above the approximate ^{9}Be(n,2n) threshold, supporting void-channel streaming and delayed solid collisions. ENDF/B-VIII.0 and FENDL-3.2c reruns preserve the same bias trend, while an all-reflective check shows that neutron return changes the partitioning of the local response. A photon-transport rerun gives a heating shift of -17.70 ± 0.17 %, close to the neutron-mode result. These are source-normalised local RVE responses, not full-blanket TBR margins.

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1. Introduction

Solid ceramic breeder blankets, in which millimetre-scale Li_4SiO_4 or Li_2TiO_3 pebbles are mixed with beryllium multiplier pebbles and cooled by helium purge gas, are central to several International Thermonuclear Experimental Reactor (ITER) test blanket module (TBM), China Fusion Engineering Test Reactor (CFETR) and demonstration fusion power plant (DEMO) blanket concepts [1, 2, 17]. Neutronic analyses of these systems commonly replace the granular breeder bed with a volume-homogenised mixture [2, 15, 17]. This approximation is attractive for full-device Monte Carlo transport because it reduces geometric complexity and makes large blanket sectors computationally tractable. However, homogenisation also removes the inter-pebble He pore network, local packing-fraction variations and discrete breeder–multiplier interfaces. These features influence the early collision history of 14.07 MeV deuterium–tritium (D–T) source neutrons. The loss of pebble-scale topology is therefore not merely a modelling convenience issue, but a neutronic accuracy issue for local ^{6}Li(n,t) tritium-production response (TPR_6), ^{9}Be(n,2n) multiplication, neutron flux and heating response.

The physical basis for this concern is well established. Random sphere packings exhibit wall-induced packing-fraction oscillations and approach their interior packing state only after several particle diameters [6, 7, 14, 16]. In a solid breeder zone, this heterogeneity changes neutron free paths, first-collision locations and the competition among threshold multiplication in Be, absorption in ^{6}Li-bearing breeder material and energy deposition in the solid phases. When the same material inventory is homogenised before solving transport, these energy- and space-dependent processes are collapsed into a single macroscopic cross section. The key issue is therefore to determine when a homogenised description preserves the relevant reaction balance and when explicit pebble geometry is required.

Previous pebble-bed neutronics studies have shown that discrete geometry can affect tritium production and local reaction distributions. For example, Zhang and Guo [4] compared homogenised, regular-packing and random-packing models for Li_2O mock-up and helium-cooled solid-breeder blanket calculations. Their random-packing model reduced the average C/E deviation of diagnostic-pellet tritium-production rates by approximately 20 %, and their Li_4SiO_4 blanket analysis showed that a small zone-integrated deviation can coexist with appreciable local deviations near pebble-bed boundaries. These studies establish the importance of explicit particle geometry, but four gaps remain for mechanism-oriented RVE analysis. First, the physical interpretation of the bias is often limited to integral tritium production, with less attention to neutron spectrum evolution, ^{9}Be(n,2n) threshold reactions, local heating and target-normalised reaction rates. Second, many comparisons use one packing or one material composition, so the sensitivity to PF, Be loading and independent packing realisations is not fully quantified. Third, the distinction between a local ^{6}Li(n,t) response, total H3-production response and full-blanket TBR is often not stated sharply enough for reviewer-proof interpretation. Fourth, the magnitude of the bias depends strongly on boundary conditions. Results obtained in leakage-dominated local models therefore cannot be transferred directly to reflected, multilayer engineering blankets without qualification.

The underlying neutronics ingredients of the present calculation are supported by external fusion-blanket benchmark studies. The Frascati Neutron Generator (FNG) HCPB test blanket module mock-up measured tritium production rates and neutron flux in a Be/Li-bearing blanket environment under 14 MeV D–T neutron irradiation, providing a reference experiment for nuclear data and transport methods in ceramic breeder blanket analyses [9]. More recent water-cooled ceramic breeder (WCCB) mock-up experiments further demonstrated tritium-production measurements in CFETR-relevant ceramic breeder blanket configurations under D–T neutron irradiation [10]. In parallel, OpenMC has been compared with MCNP, Serpent and experimental fusion benchmarks, including FNG-HCPB-type responses, with good agreement for tritium-production and activation responses in the benchmark problems considered [11]. As an executable check of the transport environment used here, we also reproduced the OKTAVIAN aluminium-sphere neutron-leakage benchmark from the SINBAD fusion benchmark suite [12, 13]. The present OpenMC calculation gave a median calculation-to-experiment (C/E) ratio of 1.213, a geometric-mean C/E of 1.046 and 85.7 % of valid energy groups within a factor of two of the measured leakage spectrum. The corresponding public reference calculations give 87.3 % for OpenMC 0.14.0 with ENDF/B-VIII.0 and 88.8 % for MCNP 6.1.1b with ENDF/B-VIII.0. This benchmark does not validate the

Li_4SiO_4/Be RVE geometry directly, but it verifies that the local OpenMC installation, D–T source treatment and source-normalised leakage-spectrum tallying reproduce a standard fusion-neutronics experiment at the same overall accuracy level as published code-benchmark calculations. These studies support the D–T source normalisation, Li-bearing tritium-production tallies, Be multiplier physics and continuous-energy Monte Carlo workflow used here, but they do not constitute a direct validation of the Li_4SiO_4/Be RVE topology. The present work therefore isolates a different question: the local homogenisation bias introduced by replacing a resolved Li_4SiO_4/Be/He pebble topology with a material-equivalent uniform medium.

The central scientific question of this work is therefore how large a local homogenisation bias is introduced when a Li_4SiO_4/Be/He pebble-bed topology is replaced by a material-equivalent uniform medium, which transport mechanism controls the bias, and how robust this bias is to controlled changes in PF, multiplier inventory and hard-sphere realisation. To address this question, we use a paired OpenMC workflow in which each representative volume element (RVE) is represented both as a homogenised material and as explicit constructive solid geometry (CSG) spheres. The two descriptions conserve the same breeder, multiplier and purge-gas inventory. Differences in TPR_6, total H3-production response, ^{9}Be(n,2n), flux and heating can therefore be attributed to geometric topology, neutron free path and collision ordering rather than to changes in material composition.

This study makes three contributions. First, it establishes a strict hard-sphere OpenMC paired-comparison protocol and uses it to quantify local homogenisation-bias ranges for the baseline composition and a 10–60 vol.% Be multiplier sweep. Second, it performs a PF sweep and independent PF=0.618 hard-sphere seed replicates to separate topology sensitivity and packing-to-packing variability from Monte Carlo tally uncertainty. Third, it uses 50-group spectra and mesh tallies to connect the global tally shifts to fast-neutron streaming, threshold Be multiplication and spatial energy-deposition redistribution. The resulting assessment route identifies when homogenisation is likely adequate and when explicit geometry or a local correction is required.

The remainder of the paper is organised as follows. Section 2 describes the homogenised and geometry-resolved OpenMC models, strict hard-sphere RVE generation, tally definitions and simulation settings. Section 3 presents the microstructural characterisation, baseline homogenised–resolved comparison, Be-fraction mechanism sweep, packing-fraction sensitivity, independent-seed repeatability and spectral/spatial diagnostics. Section 4 discusses the relation to previous pebble-bed neutronics studies, the role of boundary conditions and the engineering interpretation of the RVE results. Section 5 summarises the main conclusions and identifies the validation steps required before transferring local RVE corrections to full blanket calculations.

2. Methods

The calculation was designed as a paired comparison between two descriptions of the same material inventory: a volume-homogenised breeder-bed material and a geometry-resolved pebble bed. The audited production cases use strict non-overlapping hard-sphere geometries. This design ensures that changes in neutronic response are attributable to geometry, packing fraction or material labelling rather than to unintended changes in material inventory.

2.1. Homogenised model

The homogenised model represents the pebble bed and inter-pebble voids as a single uniform material. Its nuclide density is the volume-weighted average of the constituent solids and the He purge gas at a nominal packing fraction (PF) of approximately 0.62; the audited baseline OpenMC comparison uses PF=0.618. The density and atomic fractions follow directly from mass conservation. For each nuclide j,

$$n_j^{(\mathrm{hom})} = \mathrm{PF} \cdot \left[f_{\mathrm{Li_4SiO_4}}\, n_j^{(\mathrm{b})} + f_{\mathrm{Be}}\, n_j^{(\mathrm{Be})} \right] + (1 - \mathrm{PF})\, n_j^{(\mathrm{He})}, \tag{1}$$

where $f_{\mathrm{Li_4SiO_4}} = 0.60$ and $f_{\mathrm{Be}} = 0.40$ are the solid volume fractions of breeder and multiplier, respectively, and $n_j^{(\cdot)}$ are the nuclide densities in the corresponding pure materials.

2.2. Strict hard-sphere RVE generation

The resolved geometries used for the audited production results were generated as strict non-overlapping hard-sphere RVEs. The baseline bed contains 800 spherical pebbles in a cubic RVE. A dense parent packing was generated and then perturbed while enforcing hard-sphere non-overlap and wall clearance. The resulting particle lists were audited before OpenMC import by checking the RVE volume, total sphere volume, per-material sphere volume, packing fraction, nearest-neighbour gaps and overlap count. Production geometries with any overlapping pair were excluded from the final evidence chain.

Particles were assigned as Li_4SiO_4 breeder or Be multiplier pebbles according to the prescribed solid volume fraction. The baseline inventory used 60 vol.% breeder and 40 vol.% Be, with 90 % ^{6}Li enrichment. For the Be-fraction mechanism sweep, the particle positions and radii were kept fixed and only the breeder/Be labels were reassigned. This separates the effect of multiplier inventory from changes in packing topology. For the PF sweep, strict non-overlapping hard-sphere RVEs were generated at PF=0.56, 0.58, 0.60, 0.61 and 0.618. These cases are treated as controlled topology perturbations, not as physical thermal-expansion or irradiation-swelling calculations.

2.3. Independent realisations and geometry validity

To assess whether the PF=0.618 result was tied to one particle arrangement, three independent strict hard-sphere realisations were generated and each was transported as a paired homogenised/resolved OpenMC calculation. The independent-seed study quantifies packing-to-packing variability separately from OpenMC batchwise statistical uncertainty. All production hard-sphere particle lists used in the PF sweep and seed-repeatability study have zero overlap pairs in the pre-transport geometry audit. Direct radius-scaled thermal-expansion or swelling geometries are not used for the final physical claims unless they can be regenerated or relaxed into strictly non-overlapping geometries while conserving material inventory.

2.4. OpenMC geometry and materials

OpenMC [3] was used for all neutron-transport calculations. Each pebble in the CSV particle listing was converted into an individual `openmc.Sphere` CSG region. The inter-pebble void cell was the Boolean complement of all 800 sphere interiors within the RVE bounding box, giving a well-defined geometry of 801 cells (800 pebbles + 1 void). Boundary conditions were reflective in X and Y (infinite sheet approximation) and vacuum at $\pm Z$. The neutron source was an independent fixed source with strength 1.0 in OpenMC source-particle normalisation. Its spatial distribution was a planar box over $0 \leq x \leq 1.6\,\mathrm{cm}$ and $0 \leq y \leq 1.6\,\mathrm{cm}$ at $z = 0$; the angular distribution was monodirectional along $+z$; and the energy distribution was discrete at $14.07\,\mathrm{MeV}$, representing D–T source neutrons.

Table 1: Material definitions for the OpenMC neutron-transport calculations. The table lists the mass density, nuclide composition and temperature assigned to the Li_4SiO_4 breeder pebbles, Be multiplier pebbles and He purge-gas void phase in both the homogenised and geometry-resolved representative volume element (RVE) models. The main PF and composition sweeps used Evaluated Nuclear Data File, release B-VII.1 (ENDF/B-VII.1), data distributed by the National Nuclear Data Center (NNDC). Dedicated nuclear-data sensitivity pairs used ENDF/B-VIII.0 and FENDL-3.2c as described in Section 2.5.

Material	**Density (g/cm^3)**	**Nuclides**	T **(K)**
Li_4SiO_4 breeder	2.40	^{6}Li, ^{7}Li (90 % enriched), ^{28}Si, ^{16}O	823
Be multiplier	1.85	^{9}Be	823
He void	0.001	^{4}He	600

2.5. Tallies

Global tallies were scored in fixed-source normalisation per source neutron. The scalar tallies were `TBR_global`, using the OpenMC score `(n,t)` for ^{6}Li; `Be_n2n`, using `(n,2n)` for ^{9}Be; `heating_total`, using the OpenMC `heating` score; `flux_total`, using the track-length `flux` score; total and lithium-isotope-resolved `H3-production` tallies where available; and `dpa_total`, using `damage-energy`. The legacy tally name `TBR_global` is retained in the data files, but the reported quantity is the local RVE ^{6}Li(n,t) tritium-production response per source neutron (TPR_6), not a full-blanket design TBR. Total H3-production response is reported separately and is also a local RVE response per source neutron rather than a full-blanket TBR. OpenMC `heating` and `damage-energy` scores are native eV/source quantities; heating values are converted to MeV/source for reporting by multiplying by 10^{-6}, whereas damage-energy is retained in eV/source. The unfiltered flux tally is reported as the OpenMC track-length flux tally (particle-cm/source) and is not volume-normalised. When photon transport is enabled, an unfiltered OpenMC flux score includes all transported particle types; therefore, the photon-transport sensitivity run is used for heating-score interpretation, whereas its unfiltered flux value is not interpreted as neutron flux. Spatial tallies used an $8 \times 8 \times 8$ regular Cartesian mesh over the RVE with the same `(n,t)`, `heating`, `flux`, `(n,2n)` and `damage-energy` scores. Mesh heating is converted to MeV/source per mesh bin when reported; mesh flux is likewise a track-length tally per source particle unless explicitly divided by mesh-cell volume.

2.6. Simulation parameters and convergence

The archived and newly generated statepoints were confirmed to use OpenMC fixed-source mode rather than eigenvalue mode. The audited hard-sphere PF sweep used 80 fixed-source batches with 3×10^4 source particles per batch. The independent-seed repeatability study used 60 fixed-source batches with 2×10^4 source particles per batch. The spectrum diagnostic used 80 fixed-source batches with 3×10^4 source particles per batch. The all-reflective boundary sensitivity pair used 10 fixed-source batches with 5×10^3 source particles per batch because suppressing axial leakage substantially lengthened particle histories in the explicit RVE. The photon-transport heating-sensitivity pair used 80 fixed-source batches with 3×10^4 source particles per batch. Tally means and one-standard-deviation uncertainties are therefore batch statistics over fixed-source realisations. Some archived XML files contain legacy `inactive` values of 50 for the main RVE cases and 20 for the Be-fraction scan; these entries are not eigenvalue inactive cycles and are not associated with fission-source convergence. Photon transport was disabled in the main neutron-mode PF sweep and enabled only in the dedicated heating-sensitivity pair. The random seed was 42, and the statepoints were generated with OpenMC 0.15.0. The main RVE cases used ENDF/B-VII.1 NNDC HDF5 data. Two dedicated nuclear-data sensitivity pairs were rerun for the PF=0.618 axial-vacuum case. The first used a minimal ENDF/B-VIII.0 neutron HDF5 library [18] generated from official NNDC ENDF/B-VIII.0 source files using NJOY2016.68 [21] and OpenMC 0.15.0. This minimal library includes the nuclides required by the present model (^{6}Li, ^{7}Li, ^{9}Be, ^{16}O, ^{28}Si and ^{4}He) at 293.6, 600 and 900 K, but it does not include photon data and is therefore used only for the neutron-mode axial-vacuum sensitivity pair. The second used a FENDL-3.2c neutron ACE library [19, 20] downloaded locally from the IAEA, transferred to the remote workstation and converted to OpenMC HDF5 with `openmc_data` 2.4.0 [22] and OpenMC 0.15.0. The newly generated boundary, photon-transport, ENDF/B-VIII.0 and FENDL-3.2c runs retained captured OpenMC standard output; no lost-particle warning was found. The retained legacy PF-sweep output files contain tally summaries but not complete OpenMC standard output. The resulting relative statistical uncertainty of the RVE-level ^{6}Li(n,t) tally was 0.21–0.24 % across the main PF sweep.

2.7. Benchmark context and code verification

The present RVE is not intended to reproduce the full geometry of a specific integral benchmark. Instead, the calculation is anchored to benchmarked components of fusion blanket neutronics and then uses an internally paired comparison to isolate pebble-scale topology. The FNG-HCPB experiment provides the closest experimental analogue because it combines 14 MeV D–T source irradiation, Be multiplier material, Li-bearing breeder regions, tritium production measurements and neutron-flux diagnostics in a helium-cooled pebble bed test blanket module mock-up [9]. The WCCB D–T irradiation experiment of Zhu et al. [10] provides an additional ceramic breeder benchmark for tritium-production prediction in a CFETR-relevant blanket mock-up. At the code level, Valentine et al. [11] converted several fusion benchmark problems, including FNG-HCPB responses, to OpenMC and Serpent and compared them with established Monte Carlo workflows and available experimental data. Those results support the use of OpenMC for the source-normalised reaction-rate quantities considered here.

In addition, a local executable benchmark was run for this study using a constructive-solid-geometry reproduction of the OKTAVIAN aluminium-sphere experiment. The model consisted of a 20 cm aluminium sphere, a 0.2 cm stainless-steel shell, a central D–T source spectrum and a 134-group neutron surface-current tally at the outer boundary. The calculation used OpenMC 0.15.0, ENDF/B-VII.1 data, 50 fixed-source batches and 2×10^5 source particles per batch. After normalising the surface current to particle/source/cm^2, the 134-group leakage spectrum gave a median C/E of 1.213, a geometric-mean C/E of 1.046, 85.7 % of valid groups within a factor of two of experiment and a log-spectrum correlation coefficient of 0.823. The broad-group C/E values were 1.413 for 0.10–1 MeV, 1.184 for 1–5 MeV, 1.194 for 5–10 MeV, 0.724 for 10–15 MeV and 0.924 for 15–20.66 MeV. This result shows that the OpenMC installation and tally normalisation used in the present work reproduce the overall magnitude and energy dependence of an established 14 MeV fusion-neutronics leakage benchmark.

As a closer blanket-specific benchmark target, a reconstruction of the JAEA/FNS Li_2O pebble-bed mock-up of Sato et al. [5] was also implemented with the same OpenMC post-processing workflow. The model used the published Be/F82H/Li_2O slab dimensions, natural-Li Li_2O packing fraction of 57.8 %, two aluminium detector channels and eight detector segments per channel. Detector H3-production tallies were normalised by the number of ^{6}Li atoms in each detector segment and compared with digitised TPR curves from the published figures. With an isotropic monoenergetic D–T source, the model reproduced the axial U-shaped tritium-production profile, with shape correlations of 0.897 and 0.943 for the two detector channels, but underpredicted the absolute TPR magnitude by about one third (mean OpenMC/experiment = 0.666 and 0.670). A source-angle sensitivity calculation then bracketed the experiment: a forward-hemisphere source overpredicted the mean TPR by 34–37 %, whereas a mixed backward/forward hemisphere source with a forward-probability parameter of 0.741 gave mean calculated-to-experimental ratios of 0.968 and 1.014 for detector channels 1 and 2, respectively. The corresponding channel-wise shape correlations were 0.969 and 0.904. These results show that the present OpenMC geometry, detector tally and ^{6}Li inventory-normalisation chain can reproduce the order of magnitude and axial shape of an independent Li_2O pebble-bed TPR experiment when the source angular bias is represented. However, this remains a source-angle-calibrated screening comparison rather than a formal C/E validation, because the published paper does not provide the full FNS energy–angle source distribution and the present reconstruction still uses a homogenised, rather than explicitly repeated, Li_2O pebble-bed layer.

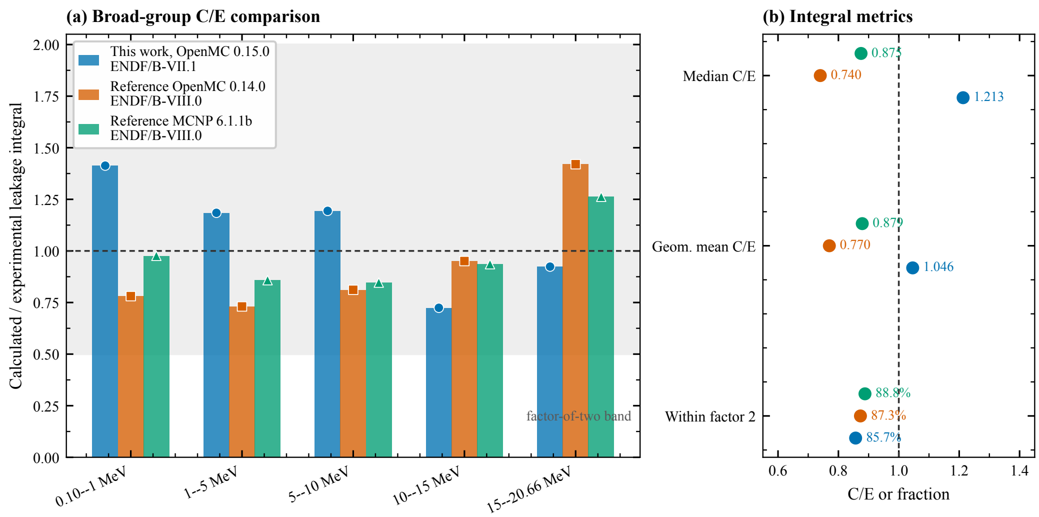

Figure 1: Comparison of the present OKTAVIAN-Al OpenMC benchmark calculation with experimental leakage data and public reference Monte Carlo results. Panel (a) compares broad-group calculated-to-experimental (C/E) ratios for the present OpenMC 0.15.0 calculation, a reference OpenMC 0.14.0 calculation and a reference MCNP 6.1.1b calculation. The grey band denotes agreement within a factor of two. Panel (b) compares integral agreement metrics for the same datasets. The reference OpenMC and MCNP data are taken from the public OKTAVIAN benchmark database used by the OpenMC Fusion Benchmarks project.

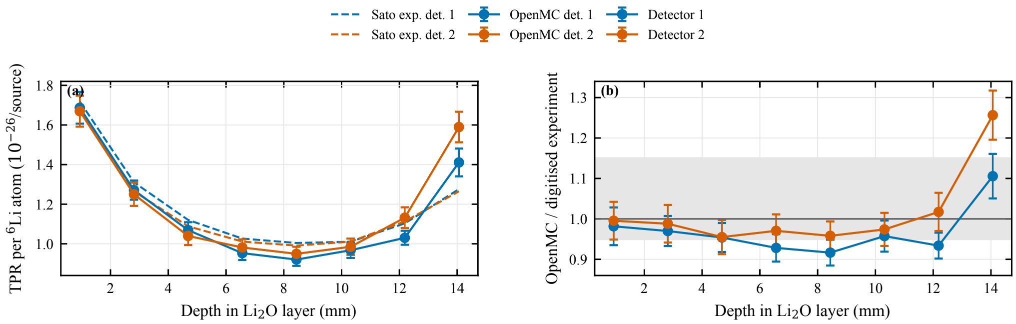

Figure 2: Source-angle-calibrated OpenMC reconstruction of the JAEA/FNS Li_2O pebble-bed mock-up compared with digitised tritium-production-rate (TPR) data from Sato et al. [5]. The OpenMC model uses the published Be/F82H/Li_2O slab dimensions, natural-Li Li_2O packing fraction of 57.8 %, two aluminium detector channels and eight detector segments per channel. Panel (a) compares detector-cell H3-production tallies normalised per ^{6}Li atom per source neutron with the digitised experimental curves. Panel (b) shows the OpenMC-to-experiment ratio for each detector segment; the grey band marks the 0.95–1.15 interval reported by Sato et al. for the original homogeneous-model C/E comparison, except at the rear boundary. Error bars denote one standard deviation (1σ) from the OpenMC detector tallies. The source angular distribution is a calibrated mixed backward/forward hemisphere approximation and is used here only as a screening benchmark, not as a replacement for the complete FNS source description.

Beyond this code-verification benchmark, the present RVE workflow was checked by four RVE-specific consistency tests. First, the homogenised and resolved cases use the same material inventory, so the reported differences are not caused by changes in Li_4SiO_4, Be or He atom densities. Second, all hard-sphere production geometries were audited for packing fraction, wall clearance and zero sphere overlap before OpenMC import. Third, tally scores, units and uncertainty propagation were audited directly from the available statepoint or raw result files. Fourth, independent PF=0.618 hard-sphere realisations reproduced the same sign and order of the resolved–homogenised shifts. Thus the external benchmarks support the transport-code and blanket-physics basis, and the Li_2O/FNS screening comparison checks the tritium-production tally normalisation against an independent pebble-bed experiment. The paired RVE audits then support the interpretation that the present conclusions arise from pebble-scale void topology rather than from inventory or numerical bookkeeping artefacts. A formal validation-grade extension would require the complete FNS energy–angle source description or an independent Li_4SiO_4/Be benchmark with published geometry, source and tally definitions.

2.8. Case matrix

Table 2 lists the calculation groups used in the audited hard-sphere analysis. The previously explored direct radius-scaling deformation cases are not used as production evidence because such geometries must first pass strict non-overlap and material-inventory-conservation audits.

Table 2: Audited simulation groups for the paired homogenised and geometry-resolved RVE calculations. PF denotes packing fraction. The table lists the main ENDF/B-VII.1 axial-vacuum production and diagnostic groups; the boundary-condition, photon-transport and nuclear-data sensitivity pairs are reported separately in Section 3.5. Each paired calculation uses the same material inventory and 14.07 MeV deuterium–tritium source. The PF sweep and independent-seed calculations use strict non-overlapping hard-sphere geometries.

Group	**Purpose**	**PF or range**
Baseline pair	Homogenised versus geometry-resolved hard-sphere RVE	0.618
PF sweep	Topology sensitivity in paired hard-sphere RVEs	0.560–0.618
Be-fraction sweep	Multiplier-inventory sensitivity at fixed topology	≈ 0.62
Seed replicates	Independent hard-sphere realisations for repeatability	0.618
Spectrum diagnostic	50-group flux comparison for mechanism analysis	0.618
Mesh diagnostic	Mid-plane spatial redistribution analysis	0.618

3. Results

3.1. The generated RVE reproduces a dense random pebble bed

The generated packing provides a reproducible microstructural reference for the neutronic comparisons. The nominal RVE reaches PF$=$ 0.6200, whereas the audited baseline transport comparison uses PF=0.618 after strict hard-sphere geometry filtering and material-inventory matching. The interior axial packing-fraction profile that is nearly uniform and boundary perturbations confined to approximately one pebble diameter at the $\pm Z$ faces (Fig. 3). The coordination-number (CN) distribution peaks at CN$=7$ with a mean of 6.6, and the specific surface area (S_v) is $S_v = 20.4\,\mathrm{cm}^{-1}$. These quantities are consistent with dense random packings of polydisperse ceramic pebbles and support the use of this RVE as a controlled local geometry rather than as an arbitrary sphere cloud.

The combination of PF, radius distribution, CN distribution and specific surface area also indicates that the selected packing is a typical dense configuration for a polydisperse ceramic pebble bed within the present parameter space. Independent random realisations with the same nominal PF, particle-size distribution and Li_4SiO_4/Be volume fractions would alter the exact locations of contacts, void throats and local first-collision sites. However, they would be expected to preserve the same statistical void topology, mean coordination environment and interfacial area scale. Such sample-to-sample differences may therefore introduce moderate stochastic fluctuations in local mesh tallies, but they are unlikely to reverse the overall trend or order of magnitude of the homogenisation bias, which is controlled by the presence of an explicit connected He void topology rather than by a particular particle arrangement.

The near-wall packing perturbation should be distinguished from the bulk-bed region sampled by the present homogenisation comparison. The disturbed layer is limited to roughly one pebble diameter near the axial boundaries, whereas most of the RVE retains a nearly constant PF close to 0.62. Its influence on the volume-integrated OpenMC tallies is therefore limited compared with the dominant contribution from the interior pebble network. Accordingly, this RVE is used to assess homogenisation bias in the approximately uniform bulk region of a pebble bed, not wall-boundary-layer behaviour.

The use of a finite 800-particle RVE is a limitation of the present study. It provides a controlled microstructural realisation for isolating the effect of explicit Li_4SiO_4/Be/He topology. To reduce the risk that the result is tied to one particle arrangement, three independent PF=0.618 strict hard-sphere realisations were also transported and compared with their matched homogenised models. Future work should extend this repeatability check to larger RVEs and, where mechanically realistic packed beds are required, to independently validated DEM or experimental particle configurations.

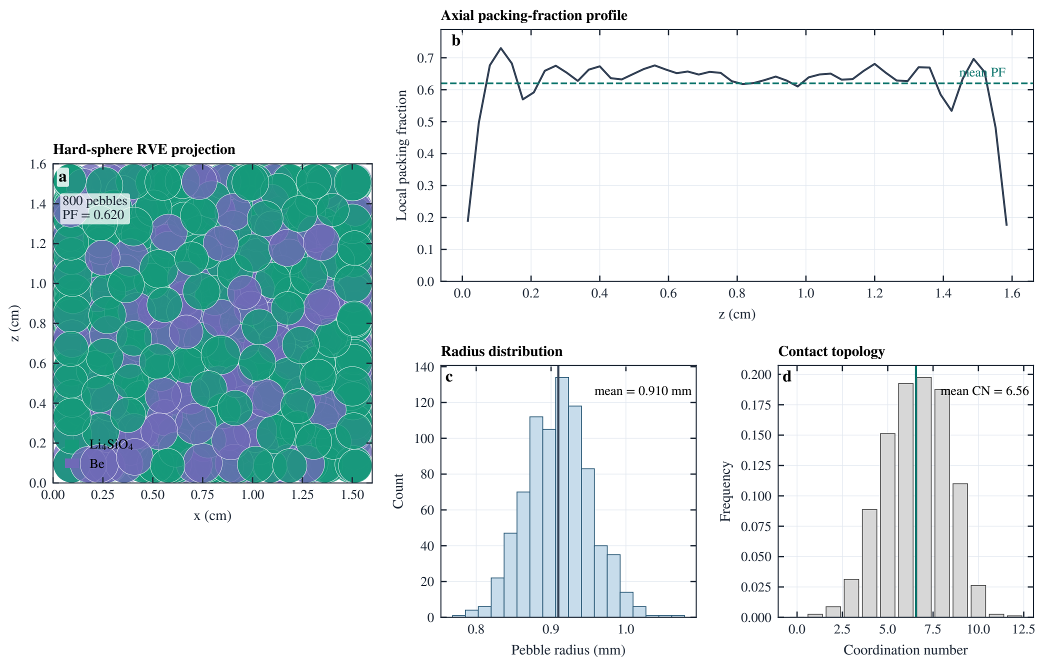

Figure 3: Strict non-overlapping hard-sphere pebble-bed microstructure used to construct the geometry-resolved RVE. The data are taken from the audited particle list exported to the OpenMC constructive solid geometry model. Panel (a) shows the x–z projection of Li_4SiO_4 breeder and Be multiplier pebbles; both spatial axes are in cm. Panel (b) shows the local axial packing fraction (PF) as a function of z (cm), with the dashed line indicating the nominal mean PF of 0.620. The audited baseline OpenMC comparison uses PF=0.618. Panel (c) shows the pebble-radius distribution in mm, and panel (d) shows the dimensionless coordination number (CN), defined as the number of mechanical contacts per pebble. These are deterministic geometry metrics; no Monte Carlo statistical uncertainty is associated with this figure.

3.2. Resolved geometry changes the global reaction balance

Replacing the homogenised bed with explicit pebbles produces a statistically significant change in all principal tallies. Relative to the material-equivalent homogenised model at PF=0.618, the resolved baseline increases TPR_6 by $+17.47 \pm 0.71$ %, total H3-production response by $+29.13 \pm 0.25$ % and ^{9}Be(n,2n) by $+45.80 \pm 0.11$ %, while decreasing the local heating tally by -18.49 ± 0.04 % and increasing the track-length flux by $+23.08 \pm 0.09$ %. These differences are substantially larger than the propagated Monte Carlo statistical uncertainties. The corresponding absolute homogenised and resolved tally values are retained in the supporting data rather than repeated as a main-text table, because the same relative shifts are shown graphically in the PF-sensitivity analysis.

The comparison is controlled by construction. All 13 material-inventory quantities, including ^{6}Li, ^{7}Li, ^{9}Be, ^{28}Si, ^{16}O and ^{4}He atom counts, solid and void volumes, and total masses, agree to < 0.02 % between the two models. The tally shifts therefore reflect the spatial arrangement of the same three-phase Li_4SiO_4/Be/He medium rather than a change in elemental inventory. In the homogenised model, the He void fraction is averaged into a uniform mixture. A 14.07 MeV D–T neutron therefore samples breeder, multiplier and purge gas through an effectively continuous macroscopic cross section. In the resolved model, the inter-pebble He phase forms connected low-collision-probability channels. These channels increase the probability that a source neutron travels several pebble diameters before its first collision, shifting the first-collision distribution toward discrete Be and breeder interfaces.

This altered collision ordering explains why the relative increase in ^{9}Be(n,2n) is much larger than the increase in TPR_6. The ^{9}Be(n,2n) reaction is driven by the fast part of the spectrum and is therefore highly sensitive to whether 14-MeV neutrons retain sufficient energy when they enter Be-rich regions. Explicit void channels preserve high-energy flight paths and increase the fast-neutron path length available for threshold multiplication in Be. TPR_6, by contrast, is governed by the integrated production of tritium in ^{6}Li after neutron multiplication, scattering and partial moderation. The ^{6}Li(n,t)^{4}He reaction samples a broader transport history than the threshold multiplication reaction. Additional secondary neutrons generated in Be must still be transported, moderated and absorbed in breeder material before contributing to TPR_6. Consequently, the Be multiplication rate responds more strongly to explicit void topology, whereas TPR_6 shows a smaller but still statistically significant increase.

The reduction in the local OpenMC heating-score tally has the same transport origin. It does not indicate a lower amount of reacting material, because the inventory is conserved. Instead, the resolved void topology redistributes energy deposition in space and energy. A larger fraction of the source-neutron population streams through He channels, undergoes multiplication or scattering at displaced Be/breeder locations, and contributes to leakage through the vacuum faces of the open RVE. Energy that would be deposited locally in the homogenised mixture is therefore partly shifted to different positions, reaction channels or leakage histories. The simultaneous increase in total flux and decrease in local heating is thus physically consistent with longer neutron residence paths and spatially displaced energy deposition, not with an inconsistency in the heating tally.

3.3. The geometry effect persists across multiplier inventory

The baseline homogenisation bias remains positive over a broad range of breeder-to-multiplier ratios. To test composition sensitivity, the explicit particle coordinates and radii were kept fixed while the solid-pebble inventory was relabelled from 10 to 60 vol.% Be. Each relabelled resolved case was paired with a homogenised case having the same total solid inventory and the same axial vacuum boundary condition. This protocol isolates the effect of void topology from the effect of changing the total amount of Be.

Across the full sweep, the resolved model enhances TPR_6 by +13.62–+17.91 % (Fig. 4). The ^{9}Be(n,2n) enhancement is larger but decreases monotonically from $+52.30 \pm 0.11$ % to $+40.05 \pm 0.07$ % as the Be inventory increases, indicating partial saturation of the multiplier response. By contrast, the total flux enhancement remains positive and slowly increases from $+21.84 \pm 0.06$ % to $+23.70 \pm 0.07$ %, whereas the OpenMC heating-score response remains lower in the resolved geometry (-21.20 ± 0.03 % to -17.33 ± 0.03 %). Thus, the resolved-geometry effect is not a single-composition anomaly, but a systematic transport effect whose partitioning among breeding, multiplication and heating depends on multiplier inventory.

The monotonic decrease in the relative ^{9}Be(n,2n) enhancement reflects the evolution of the fast-neutron field as the multiplier fraction increases. The ^{9}Be(n,2n) channel is a threshold reaction controlled by the portion of the spectrum that remains above the reaction threshold after 14.07 MeV source neutrons stream through He voids and scatter in the solid pebbles. At low Be fraction, the homogenised model dilutes Be uniformly through the RVE. The resolved model instead allows high-energy neutrons to stream through connected He pores and intersect discrete Be pebbles before substantial energy degradation. Explicit geometry therefore produces a large relative increase in the probability that a fast neutron samples a Be target while it still has sufficient energy for multiplication. As the Be fraction increases, both the homogenised and resolved models already provide a high probability of Be encounter along a typical neutron trajectory. The additional benefit of explicit void topology then becomes smaller in relative terms, even though the absolute multiplication probability increases with Be inventory.

This behaviour indicates multiplier-reaction saturation rather than a loss of the geometric effect. Increasing the number of Be pebbles raises the material concentration available for ^{9}Be(n,2n), but it also increases competition among neighbouring Be regions for the same population of uncollided or weakly moderated fast neutrons. Once streamed neutrons are already likely to collide with Be within their first few collisions, additional Be mainly replaces breeder or He space along trajectories that have partly lost energy. These later collisions contribute less efficiently to the threshold reaction because the spectrum has been softened by elastic and inelastic scattering. Resolved pebble contacts and finite Be domains also introduce a self-shielding-like effect. The outer portions of Be pebbles preferentially intercept the fast-neutron current, whereas inner or downstream Be volume is sampled after attenuation and energy degradation. Thus, the material-concentration effect increases the available number of Be target atoms, while the geometry effect controls how much high-energy path length is exposed to those targets. The decreasing relative gain from +52.30 % to +40.05 % indicates that the latter effect approaches a saturated regime as the accessible fast-neutron current becomes the limiting quantity.

The coupled trends in TPR_6 and flux are consistent with this interpretation. The total flux enhancement remains nearly constant over the Be-fraction sweep, showing that the principal topological change is the redistribution of neutron free paths by the connected He void topology rather than a strong change in the overall neutron population. The reaction-specific responses then depend on how this redistributed flux samples energy-dependent cross sections. The ^{9}Be(n,2n) response decreases in relative magnitude because the threshold multiplier channel is limited by the availability of fast neutrons in Be-rich regions. TPR_6 varies more weakly because tritium production integrates multiplication, scattering, moderation and absorption in ^{6}Li-bearing Li_4SiO_4 pebbles. Secondary neutrons generated in Be can increase the neutron supply to the breeder, but the concurrent reduction in breeder fraction and the requirement for transport into Li_4SiO_4 moderate the TPR_6 response. The Be-fraction sweep therefore separates a material-inventory effect, which changes the number of multiplier targets, from a persistent topology effect, which controls neutron streaming, first-collision ordering and the exposure of Be to the fast part of the D–T spectrum.

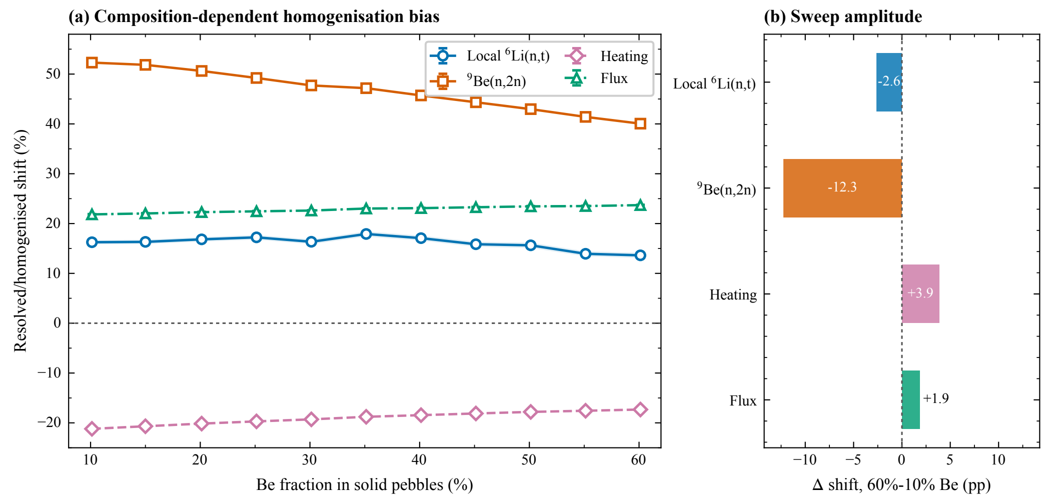

Figure 4: Be multiplier-fraction sweep used to evaluate the composition dependence of the homogenisation bias. Each point compares a geometry-resolved RVE with a matched homogenised RVE having the same total material inventory, the same 14.07 MeV source and the same axial vacuum boundary condition. Panel (a) shows the resolved–homogenised relative shift, in percent, for TPR_6, ^{9}Be(n,2n), OpenMC heating-score response and total flux as functions of the Be fraction in the solid pebble inventory (vol.%). Panel (b) gives the change in each relative shift between the 10 % and 60 % Be cases, expressed in percentage points (pp), to quantify the sweep amplitude. Error bars denote one standard deviation (1σ) propagated from the paired OpenMC tallies; all tallies use ENDF/B-VII.1 cross-section data.

3.4. Packing-fraction sensitivity in strict hard-sphere RVEs

Packing-fraction sensitivity was evaluated with strict non-overlapping hard-sphere RVEs at PF values of 0.56, 0.58, 0.60, 0.61 and 0.618. This sweep is interpreted as a controlled topology perturbation rather than as a thermal-expansion or irradiation-swelling model. For each PF, the homogenised and resolved calculations use the same material inventory and source/boundary conditions, so the resolved–homogenised difference remains a topology-induced bias.

The PF sweep shows a systematic attenuation of the resolved-geometry enhancement as the packing becomes denser (Fig. 5). The TPR_6 shift decreases from $+22.92 \pm 0.85$ % at PF=0.56 to $+17.47 \pm 0.71$ % at PF=0.618. The total H3-production response shift decreases from $+34.88 \pm 0.29$ % to $+29.13 \pm 0.25$ %, and the ^{9}Be(n,2n) shift decreases from $+51.67 \pm 0.11$ % to $+45.80 \pm 0.11$ %. The heating shift weakens from -21.99 ± 0.04 % to

-18.49 ± 0.04 %, whereas the flux shift remains positive and nearly constant, changing from $+22.39 \pm 0.09$ % to $+23.08 \pm 0.09$ %. These trends indicate that the explicit-geometry effect is strongest when connected He void paths occupy a larger fraction of the RVE, because fast neutrons can stream farther before sampling solid material. As PF increases, the void network narrows and the first-collision distribution of the resolved model moves closer to the volume-averaged collision probability of the homogenised model. The homogenisation bias therefore decreases but does not vanish at dense packing.

This PF dependence supports a topology-saturation interpretation for multiplier reactions. Increasing PF raises the solid collision probability and reduces the relative importance of long He streaming paths. However, the resolved ^{9}Be(n,2n) enhancement remains larger than the TPR_6 enhancement throughout the sweep because the multiplier reaction remains more sensitive to the local fast-neutron exposure of Be surfaces. The flux shift is comparatively insensitive to PF because the track-length tally records path-length redistribution over the whole RVE, whereas TPR_6, H3 production, ^{9}Be(n,2n) and heating depend on where this redistributed flux samples energy-dependent reaction channels.

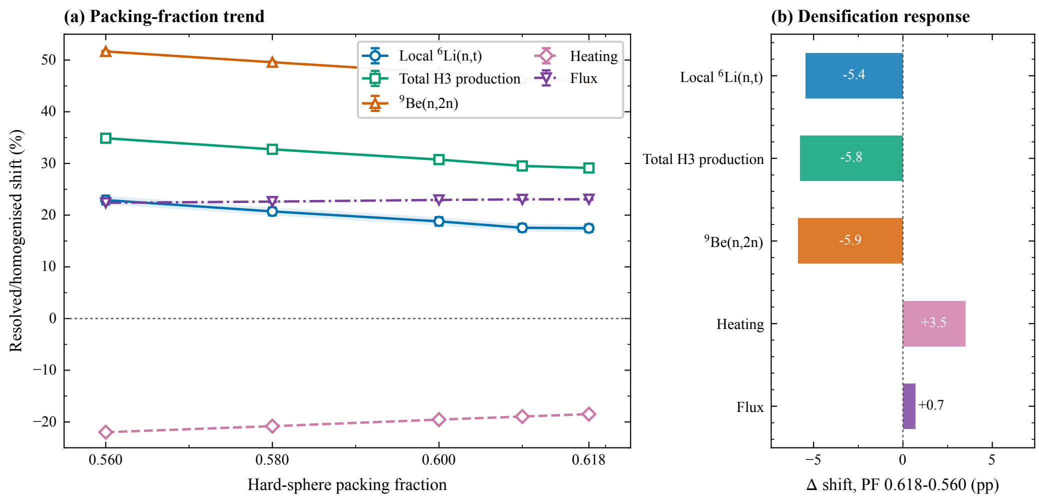

Figure 5: Packing-fraction dependence of resolved–homogenised tally shifts in strict non-overlapping hard-sphere RVEs. Each point represents a paired homogenised/resolved OpenMC calculation with the same material inventory, 14.07 MeV source, reflective lateral boundaries, axial vacuum boundaries and ENDF/B-VII.1 neutron data. Panel (a) shows the relative differences $(B/A-1)\times 100$ % for TPR_6, total H3-production response, ^{9}Be(n,2n), heating and track-length flux. Panel (b) gives the change in each relative shift between PF=0.560 and PF=0.618, expressed in percentage points (pp), to highlight the densification response. Error bars denote propagated one-standard-deviation (1σ) tally uncertainties.

3.5. Boundary-condition, heating-score and nuclear-data sensitivity

The PF=0.618 baseline was extended with three targeted sensitivity checks. First, the axial vacuum faces were replaced by reflective boundaries while keeping the same hard-sphere particle list, material inventory and ENDF/B-VII.1 data. In this all-reflective local RVE, the resolved–homogenised TPR_6 shift decreases from $+17.47\pm0.71$ % to $+15.57\pm0.72$ %, and the total H3-production response shift decreases from $+29.13\pm0.25$ % to $+15.91\pm0.71$ %. The reduction confirms that axial vacuum leakage amplifies part of the local breeding response. However, the ^{9}Be(n,2n) shift remains large ($+61.52 \pm 0.87$ %), showing that the threshold multiplier response is still controlled by the exposure of discrete Be surfaces to fast neutrons even when leakage is suppressed. The all-reflective result should be interpreted as a local sensitivity calculation rather than a blanket-scale reflected-boundary design value, because the closed RVE also increases particle residence time and does not include structural attenuation or multilayer blanket leakage paths.

Second, the PF=0.618 axial-vacuum pair was rerun with photon transport enabled. The coupled photon-transport run gives a heating shift of -17.70 ± 0.17 %, close to the neutron-mode heating shift of -18.49 ± 0.04 %. This agreement indicates that the sign and approximate magnitude of the local heating redistribution are not artifacts of disabling photon transport in the main neutron-mode calculations. The reaction-rate shifts in the photon run also remain consistent with the neutron-mode baseline, with TPR_6, H3 production and ^{9}Be(n,2n) shifts of $+17.16 \pm 0.70$ %, $+29.02 \pm 0.25$ % and $+45.75 \pm 0.11$ %, respectively. The unfiltered flux tally from the photon-transport run is retained in the source CSV for reproducibility but is not used as a neutron-flux metric because OpenMC includes all transported particle types in an unfiltered flux score when photon transport is enabled.

Third, the PF=0.618 axial-vacuum pair was repeated with generated ENDF/B-VIII.0 and FENDL-3.2c neutron libraries. The resolved–homogenised shifts remain consistent with the ENDF/B-VII.1 baseline. With ENDF/B-VIII.0, TPR_6 changes from $+17.47 \pm 0.71$ % to $+18.00 \pm 0.74$ %, total H3 production from $+29.13 \pm 0.25$ % to $+29.32 \pm 0.26$ %, ^{9}Be(n,2n) from $+45.80 \pm 0.11$ % to $+46.11 \pm 0.11$ %, heating from -18.49 ± 0.04 % to -19.01 ± 0.04 % and track-length flux from $+23.08 \pm 0.09$ % to $+23.14 \pm 0.09$ %. With FENDL-3.2c, the corresponding shifts are $+17.73 \pm 0.73$ %, $+29.22 \pm 0.26$ %, $+45.78 \pm 0.11$ %, -18.64 ± 0.04 % and $+23.09 \pm 0.09$ %. The sign and magnitude of the topology-induced local bias are therefore not specific to the ENDF/B-VII.1 evaluation for the nuclides and neutron-mode responses considered here. These results should be interpreted as targeted neutron-data checks, not as a complete nuclear-data uncertainty analysis.

3.6. Independent hard-sphere realisations give repeatable bias trends

The representativeness of the PF=0.618 hard-sphere result was tested with three independent non-overlapping realisations. The repeatability check separates packing-to-packing variability from OpenMC tally uncertainty and directly addresses the limitation of using a single RVE realisation.

The mean resolved–homogenised shifts over the three realisations are +16.78 % for TPR_6, +28.74 % for total H3-production response, +45.69 % for ^{9}Be(n,2n), −18.52 % for heating and +23.08 % for flux (Fig. 6). The corresponding sample standard deviations are 0.64, 0.28, 0.07, 0.04 and 0.13 percentage points, respectively. The sign and magnitude of the main homogenisation bias are therefore robust to independent hard-sphere realisations within the tested parameter set. This does not replace a full RVE-

Table 3: PF=0.618 sensitivity summary for paired homogenised and geometry-resolved RVE calculations. All values are resolved–homogenised relative shifts, $(B/A - 1) \times 100$ %, with propagated one-standard-deviation (1σ) statistical uncertainty. TPR_6 is the local ^{6}Li(n,t) tritium-production response per source neutron. The axial-vacuum ENDF/B-VII.1, ENDF/B-VIII.0, FENDL-3.2c and photon-transport rows used 80 fixed-source batches with 30000 particles per batch. The all-reflective case used 10 fixed-source batches with 5000 particles per batch because the absence of axial leakage substantially lengthened resolved-geometry particle histories. The photon-transport row is used for heating-score interpretation; its unfiltered flux tally contains both neutron and photon track-length contributions and is not interpreted as neutron flux.

Case	**TPR_6**	**H3 production**	**^{9}Be(n,2n)**	**Heating**	**Flux**
Axial vacuum, neutron mode	$+17.47 \pm 0.71$	$+29.13 \pm 0.25$	$+45.80 \pm 0.11$	-18.49 ± 0.04	$+23.08 \pm 0.09$
Axial vacuum, ENDF/B-VIII.0	$+18.00 \pm 0.74$	$+29.32 \pm 0.26$	$+46.11 \pm 0.11$	-19.01 ± 0.04	$+23.14 \pm 0.09$
Axial vacuum, FENDL-3.2c	$+17.73 \pm 0.73$	$+29.22 \pm 0.26$	$+45.78 \pm 0.11$	-18.64 ± 0.04	$+23.09 \pm 0.09$
All-reflective, neutron mode	$+15.57 \pm 0.72$	$+15.91 \pm 0.71$	$+61.52 \pm 0.87$	-1.89 ± 0.38	$+130.97 \pm 3.24$
Axial vacuum, photon transport	$+17.16 \pm 0.70$	$+29.02 \pm 0.25$	$+45.75 \pm 0.11$	-17.70 ± 0.17	$+23.30 \pm 0.12$[a]

[a]Unfiltered OpenMC flux with photon transport enabled includes neutron and photon contributions; this value is retained in the source CSV but is not used as a neutron-flux metric.

convergence study, but it shows that the principal mechanism is not caused by one particular particle arrangement.

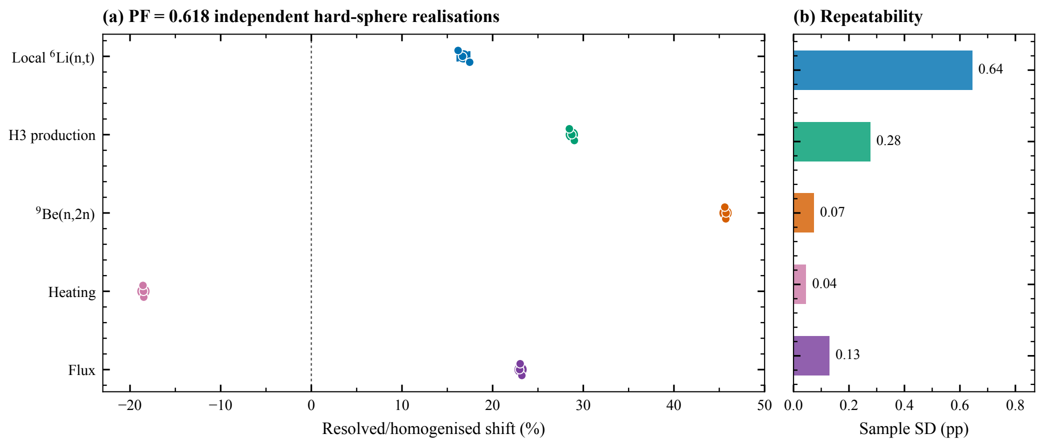

Figure 6: Resolved–homogenised tally shifts for three independent PF=0.618 hard-sphere RVE realisations. Panel (a) shows the relative shift for each independent packing, with the horizontal ranges indicating the min–max spread and the mean marker with sample-standard-deviation error bar. Panel (b) summarises the packing-to-packing sample standard deviation in percentage points (pp). This variability is distinct from OpenMC batch-wise tally uncertainty.

3.7. Spectral and spatial diagnostics support the transport mechanism

The 50-group spectrum diagnostic links the tally shifts to neutron-energy redistribution. In the PF=0.618 case, the resolved/homogenised flux ratio integrated over the 0.5 eV–0.1 MeV window is 1.520, and the corresponding ratio for 0.1–1.85 MeV is 1.549 (Fig. 7). Above the approximate ^{9}Be(n,2n) threshold of 1.85 MeV, the integrated ratio is 1.221. Thus, the explicit geometry increases not only the total track-length flux but also the population of neutrons in the fast and intermediate energy ranges that feed multiplier reactions and subsequent breeder absorption.

Qualitatively, the homogenised spectrum and the resolved spectrum represent two different moderation pathways for the same 14.07 MeV D–T source. In the homogenised model, the source neutron continuously samples the volume-averaged Li_4SiO_4/Be/He macroscopic cross section, so scattering, threshold multiplication and absorption are mixed over a shortened effective transport scale. In the resolved model, inter-pebble He channels provide low-density flight paths that delay the first solid collision and retain a larger fast-neutron component near discrete Be interfaces. This harder local fast-neutron exposure directly supports the higher ^{9}Be(n,2n) rate, while secondary neutrons produced in Be subsequently increase the neutron supply available for ^{6}Li-bearing breeder pebbles. The threshold-window increase therefore provides direct mechanistic support for the observed combination of enhanced TPR_6, enhanced H3-production response, enhanced multiplier reaction rate, increased flux and reduced local heating.

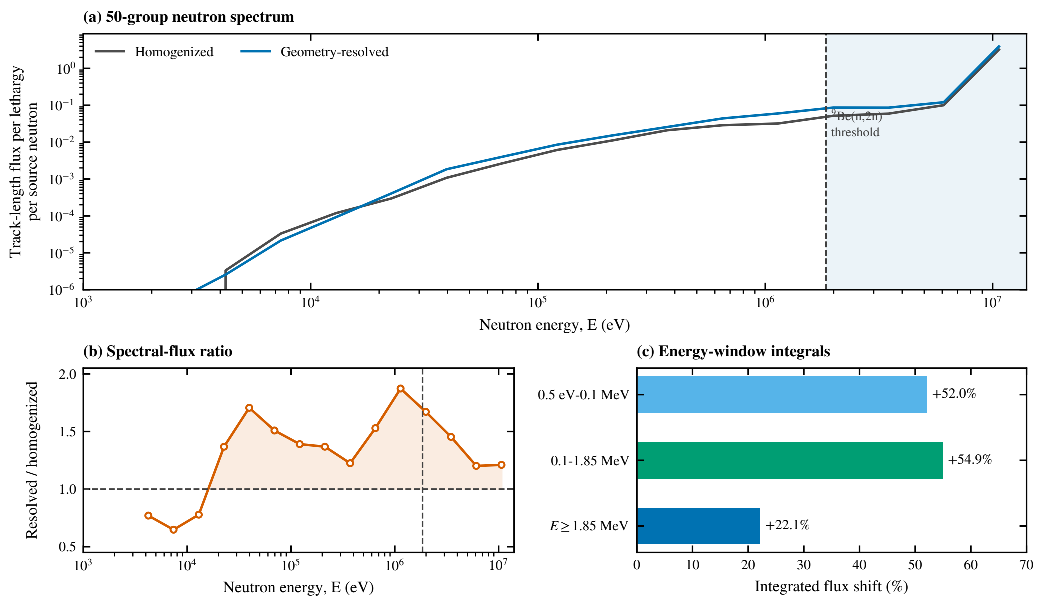

Figure 7: Fifty-group neutron spectrum diagnostic for the PF=0.618 hard-sphere RVE. Panel (a) shows the OpenMC track-length flux per lethargy per source neutron for the homogenised and geometry-resolved models. Panel (b) shows the resolved/homogenised spectral-flux ratio for energy bins above 1 keV. Panel (c) summarises the integrated resolved–homogenised flux shift in selected energy windows. The vertical dashed line marks the approximate ^{9}Be(n,2n) threshold at 1.85 MeV. The data are obtained from paired fixed-source OpenMC calculations with the same material inventory, source and boundary conditions.

Mid-plane mesh diagnostics show that the resolved geometry redistributes local reaction rates within the RVE. In the PF=0.618 mid-plane, the mean mesh-bin shifts are positive for TPR_6 and H3 production, negative for heating and weakly positive for flux, but individual bins exhibit large spatial variation because local material placement, void connectivity and tally statistics vary at the cell scale. These maps should therefore be interpreted as evidence of spatial redistribution rather than as a deterministic correction field. Their main role is to confirm that the global shifts arise from heterogeneous transport paths through the explicit Li_4SiO_4/Be/He topology.

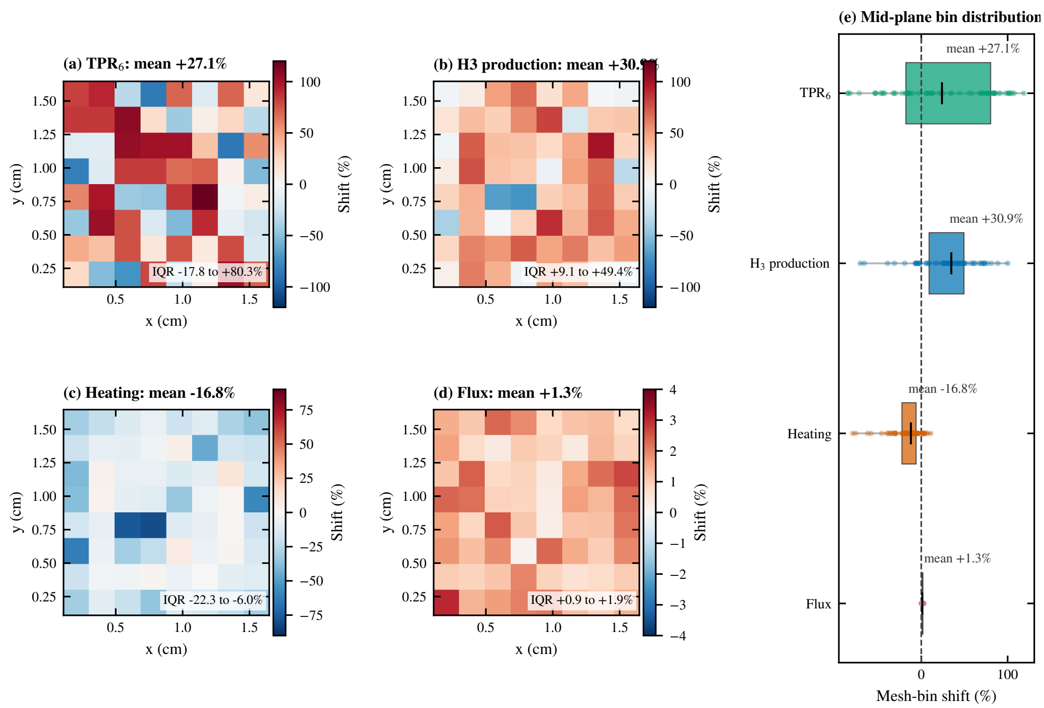

Figure 8: Mid-plane mesh-bin resolved–homogenised shifts for the PF=0.618 hard-sphere RVE. The maps are extracted from the $8 \times 8 \times 8$ Cartesian OpenMC mesh at the RVE mid-plane. Panels show the cell-wise relative shifts for (a) TPR$_6$, (b) total H3-production response, (c) heating and (d) track-length flux; panel (e) gives the corresponding distribution of mid-plane mesh-bin shifts, with boxes denoting the interquartile ranges and vertical ticks denoting the medians. Spatial axes are x and y in cm. Colour bars denote $(B/A-1) \times 100$ % for each mesh bin; mesh-bin statistical uncertainties are not plotted to preserve map readability.

4. Discussion

The central result of this study is that homogenisation can change the local reaction balance of a breeder-bed RVE even when material inventory is conserved to high precision. The effect is not limited to TPR$_6$. ^{9}Be(n,2n), total flux and local heating change in different directions, indicating a transport-history effect rather than a simple material-density effect. The physical origin is the replacement of a connected three-phase void topology by a uniform macroscopic cross section in the homogenised model. This distinction matters for coupled blanket analyses because thermal, swelling and damage calculations often use the same Monte Carlo tallies that are affected by homogenisation.

4.1. Relation to previous pebble-bed neutronics studies

The positive TPR_6 shift found here differs from the small negative zone-integrated Li-6 tritium-production deviation reported by Zhang and Guo [4] for a reflected helium-cooled blanket. In their Li_4SiO_4 region-1 analysis, the random-packing local-bin deviations relative to homogenisation ranged from -8.05 % to +2.41 %, whereas the integrated deviation was only -0.15 %. This difference should not be read as a contradiction. Reflected-neutron effects are known to influence fusion-blanket tritium-production interpretation [8], and the two calculations isolate different parts of the problem. Four modelling choices are particularly important:

1. **Boundary conditions.** Our vacuum $\pm Z$ allows streaming neutrons to exit, maximising the streaming effect; their reflective azimuthal boundaries and Be reflector layers recirculate neutrons, reducing net enhancement.
2. **Packing-fraction distribution.** Our Lubachevsky–Stillinger (LS) algorithm gives near-constant PF ($\approx$0.62) throughout; their random packing exhibits $\pm$8 % near-wall PF oscillations that drive compensating negative local deviations.
3. **Material composition.** Our model includes 40 vol% Be multiplier, whose (n,2n) reactions thermalise the spectrum and increase ^{6}Li absorption; their Section 4 uses pure Li_4SiO_4.
4. **Source.** Monoenergetic vs. Gaussian-distributed.

The present calculation is therefore a controlled local homogenisation test, whereas Zhang and Guo address a more blanket-like reflected configuration. Together, the two studies indicate that explicit pebble geometry can remain locally important even when a full-blanket global tritium-production shift is small.

4.2. Boundary-condition scope and engineering interpretation

The vacuum boundaries on the axial z faces create a strong leakage channel and amplify free-streaming effects through the connected He void topology. In this configuration, a 14.07 MeV D–T neutron entering an inter-pebble void can travel farther before collision, reach discrete Be or Li_4SiO_4 interfaces with less spectral degradation, or leave the RVE through the vacuum faces. This boundary choice isolates the homogenisation error associated with replacing the explicit three-phase topology by a uniform macroscopic cross section. The absolute homogenised–resolved differences reported here should therefore be regarded as leakage-sensitive upper-bound estimates for this local RVE, not as direct full-blanket design margins.

The all-reflective sensitivity calculation confirms that this boundary effect is not merely qualitative. Suppressing axial leakage reduces the TPR_6 shift from $+17.47 \pm 0.71$ % to $+15.57 \pm 0.72$ % and reduces the total H3-production shift from $+29.13 \pm 0.25$ % to $+15.91 \pm 0.71$ %. The local heating shift also weakens from -18.49 ± 0.04 % to -1.89 ± 0.38 %, indicating that neutron return partially compensates the leakage-driven reduction in local energy deposition. By contrast, the ^{9}Be(n,2n) shift increases to $+61.52 \pm 0.87$ %. This behaviour is physically plausible for a closed local RVE: reflected fast and epithermal histories increase the residence time and allow repeated sampling of discrete Be surfaces, while the homogenised model still distributes Be continuously through the volume. The result therefore shows that boundary conditions can change not only the magnitude but also the partitioning of the homogenisation bias among breeding, multiplication, flux and heating responses.

The engineering implication is therefore not that the numerical shifts in this RVE can be inserted directly into a solid breeder blanket design. Rather, the RVE provides a controlled upper-bound test of the mechanism by which pebble-bed topology changes neutron free path, collision ordering, spectral degradation and reaction-site distribution. For reflected or multi-layer blanket calculations, the appropriate transfer of this result is to retain the mechanism, not the absolute percentage values. A practical workflow is to use explicit geometry-resolved RVEs as local correction or uncertainty-quantification models inside representative blanket zones, repeat the comparison under the actual module boundary condition or response-matrix environment, and then propagate the resulting local homogenisation bias to full-sector Monte Carlo calculations. Such an approach would allow the effects identified here, namely void-channel streaming, fast-neutron exposure of Be, breeder absorption after moderation and spatial redistribution of nuclear heating, to be evaluated under realistic reflective neutron return, coolant-channel shielding and structural-material attenuation.

The PF and composition sweeps are parametric local sensitivity studies. They show that the sign and order of the local homogenisation bias persist under controlled hard-sphere topology and Be-inventory perturbations, but they do not constitute a validated thermal-expansion, irradiation-swelling or full-blanket feedback model. The absolute RVE-level TPR_6, the magnitude of leakage-driven spatial peaks and any architecture-specific breeding margin remain outside the claims of this paper.

On this basis, a reflective or multilayer blanket environment should not be represented by directly transferring the axial-vacuum percentages. The new all-reflective RVE check demonstrates sensitivity to neutron return, but it is still a closed local cell rather than a multilayer blanket segment with structural attenuation, coolant channels and finite leakage. A modern engineering application therefore requires reflected or multilayer OpenMC calculations with the relevant blanket architecture before any quantitative correction is applied to design.

4.3. Engineering use of the homogenised model

The present results can be translated into a practical screening rule for engineering neutronics, provided that the rule is applied to local modelling choices rather than to the final blanket breeding margin. In regions where a paired resolved–homogenised comparison, or an experimentally validated local correction, indicates a TPR_6 deviation below about 5 %, the homogenised model may be acceptable for preliminary design studies, global optimisation and parametric screening, especially when the quantity of interest is an integral blanket response. This threshold is not a universal safety criterion. It is a pragmatic engineering tolerance below which the geometry-induced uncertainty is comparable to other modelling uncertainties in early blanket analysis. Conversely, the present RVE gives TPR_6 deviations of +13.62–+17.91 % across the Be-fraction sweep, together with much larger ^{9}Be(n,2n) deviations and a substantial heating redistribution. A deviation of this order should be treated as evidence that explicit pebble geometry, or a locally calibrated correction factor derived from resolved RVEs, is required before the result is used in design-sensitive tritium-production or nuclear-heat assessments.

The model choice should also depend on the structural location within the blanket. In the approximately uniform bulk region of a pebble bed, where the packing fraction is close to its interior value and large void clusters are absent, homogenisation can remain useful if the local bias is demonstrated to fall within the acceptable range defined above. Near-wall boundary layers, purge-gas gaps, coolant-channel neighbourhoods, local packing defects and regions with concentrated void connectivity require a more conservative treatment. In these zones, neutron streaming, first-collision ordering and spectral hardening are governed by geometry rather than by the volume-averaged material composition, so explicit geometry or a resolved-RVE-based correction should be used for TPR_6, $^9Be(n,2n)$, flux and heating tallies.

The operating state provides a second screening dimension. For a static pebble bed with a stable packing fraction and no significant irradiation-induced redistribution, a homogenised model may be adequate after a one-time resolved-RVE verification of the local bias. If future thermal-expansion, uniform-swelling or flux-correlated-swelling cases are used, they should first be regenerated as non-overlapping geometries with material-inventory-conserving density scaling. Until such calculations are available, the present PF sweep should be interpreted only as a controlled hard-sphere topology sensitivity study.

Accordingly, the RVE results should be used as a correction basis and a decision framework for full-blanket calculations, not as direct design values. A recommended workflow is to perform full-sector calculations with homogenised materials, identify breeder-bed zones that are close to geometry-sensitive conditions, compute representative resolved RVEs for those zones under the actual material composition and boundary environment, and propagate the resulting local correction or uncertainty into the full model. This workflow preserves the computational efficiency of homogenised neutronics while avoiding unqualified use of homogenisation in regions where pebble-scale topology controls multiplication, breeding or energy deposition.

5. Conclusions and outlook

This work demonstrates that explicit pebble-scale geometry provides a first-order local correction to homogenised neutronics in a strict hard-sphere Li_4SiO_4/Be breeder-bed RVE. With identical material inventory at PF=0.618, the resolved baseline increases TPR_6 by $+17.47 \pm 0.71$ %, total H3-production response by $+29.13 \pm 0.25$ % and $^9Be(n,2n)$ by $+45.80 \pm 0.11$ %, while reducing the OpenMC heating score by -18.49 ± 0.04 % and increasing track-length flux by $+23.08 \pm 0.09$ %. The hard-sphere PF sweep and Be-fraction sweep show that the bias persists under controlled topology and multiplier-inventory perturbations, and the independent-seed study confirms that the sign and order of the shifts are not tied to a single particle arrangement.

The mechanism is controlled by connected He void topology. Explicit void paths delay solid collisions, preserve fast-neutron exposure to Be, enhance threshold ^{9}Be(n,2n) multiplication and redistribute local energy deposition. The 50-group spectrum diagnostic supports this interpretation by showing a resolved/homogenised flux ratio of 1.221 above the approximate ^{9}Be(n,2n) threshold. The ENDF/B-VIII.0 and FENDL-3.2c sensitivity runs preserve the same sign and approximate magnitude of the axial-vacuum local bias. The ENDF/B-VIII.0 shifts are $+18.00 \pm 0.74$ % for TPR_6, $+29.32 \pm 0.26$ % for total H3 production, $+46.11 \pm 0.11$ % for ^{9}Be(n,2n), -19.01 ± 0.04 % for heating and $+23.14 \pm 0.09$ % for flux. The corresponding FENDL-3.2c shifts are $+17.73 \pm 0.73$ %, $+29.22 \pm 0.26$ %, $+45.78 \pm 0.11$ %, -18.64 ± 0.04 % and $+23.09 \pm 0.09$ %. The all-reflective sensitivity check reduces the TPR_6 and H3-production shifts to $+15.57 \pm 0.72$ % and $+15.91 \pm 0.71$ %, respectively, while increasing the ^{9}Be(n,2n) shift to $+61.52 \pm 0.87$ %; this confirms that boundary conditions modify the partitioning of the local bias. A photon-transport rerun gives a heating shift of -17.70 ± 0.17 %, close to the neutron-mode value, supporting the robustness of the heating-redistribution interpretation. The reported quantities are source-normalised local RVE responses, not full-blanket TBR margins. Future work should extend the same audited workflow to multilayer blanket environments and validated microstructure-evolution geometries that conserve material inventory and remain strictly non-overlapping.

Data and code availability

The input particle lists, OpenMC input-generation scripts, post-processing scripts, processed tally CSV files, figure source data and manuscript build files are available at `https://github.com/wangjianfttt/fusion-pebble-bed-rve-openmc-data` and archived on Zenodo at `https://doi.org/10.5281/zenodo.20636518`. Continuous-energy nuclear-data libraries are not redistributed because of their size and licensing conditions; the exact library names, source URLs, processing tools, checksums and local `cross_sections.xml` paths used in the calculations are recorded in the data package.

CRediT authorship contribution statement

Jian Wang: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft. Haoxi Wang: Methodology, Writing – review and editing. Zhiyuan Liu: Application context, Methodology, Writing – review and editing. Haishun Deng: Application context, Validation planning, Writing – review and editing. Wei Wen: Data curation, Visualization, Writing – review and editing. Gang Shen: Supervision, Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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